

As explained in § 3.1 ‘*Payment Addresses and Keys*’ on p. 14, Sapling allows the efficient creation of many *diversified payment addresses* with the same *spending authority*. A group of such addresses shares the same *full viewing key* and *incoming viewing key*.

To create a new *diversified payment address* given an *incoming viewing key* ivk , repeatedly pick a *diversifier* d uniformly at random from $\mathbb{B}^{[\ell_d]}$ until the *diversified base* $g_d = \text{DiversifyHash}^{\text{Sapling}}(d)$ is not $\perp$. Then calculate the *diversified transmission key* pk_d :

$$pk_d := KA^{\text{Sapling}}.\text{DerivePublic}(ivk, g_d).$$

The resulting *diversified payment address* is $(d : \mathbb{B}^{[\ell_d]}, pk_d : KA^{\text{Sapling}}.\text{PublicPrimeOrder})$.

For each *spending key*, there is also a *default diversified payment address* with a “random-looking” *diversifier*. This allows an implementation that does not expose diversified addresses as a user-visible feature, to use a default address that cannot be distinguished (without knowledge of the *spending key*) from one with a random *diversifier* as above. Note however that the *zcashd* wallet picks *diversifiers* as in [ZIP-32], rather than using this procedure.

Let $\text{first} : (\mathbb{B}^{\mathbb{Y}} \rightarrow T \cup \{\perp\}) \rightarrow T \cup \{\perp\}$ be as defined in § 5.4.9.5 ‘*Group Hash into Jubjub*’ on p. 107. Define:

$$\text{CheckDiversifier}(d : \mathbb{B}^{[\ell_d]}) := \begin{cases} \perp, & \text{if } \text{DiversifyHash}^{\text{Sapling}}(d) = \perp \\ d, & \text{otherwise} \end{cases}$$

$$\text{DefaultDiversifier}(sk : \mathbb{B}^{[\ell_{sk}]}) := \text{first}(i : \mathbb{B}^{\mathbb{Y}} \mapsto \text{CheckDiversifier}(\text{truncate}_{(\ell_d/8)}(\text{PRF}_{sk}^{\text{expand}}([3, i]))) : \mathbb{J}^{(r)*} \cup \{\perp\}).$$

For a random *spending key*, $\text{DefaultDiversifier}$ returns $\perp$ with probability approximately 2^{-256} ; if this happens, discard the key and repeat with a different sk .

Notes:

- The protocol does not prevent using the *diversifier* d to produce “vanity” addresses that start with a meaningful string when encoded in *Bech32* (see § 5.6.3.1 ‘*Sapling Payment Addresses*’ on p. 119). Users and writers of software that generates addresses should be aware that this provides weaker privacy properties than a randomly chosen *diversifier*, since a vanity address can obviously be distinguished, and might leak more information than intended as to who created it.
- Similarly, address generators **MAY** encode information in the *diversifier* that can be recovered by the recipient of a payment to determine which *diversified payment address* was used. It is **RECOMMENDED**, instead of directly encoding information in the *diversifier*, to encode it in the *diversifier index* specified in [ZIP-32]. This ensures that the information is only accessible to a holder of the *diversifier key* dk .

Non-normative notes:

- Assume that $\text{PRF}^{\text{expand}}$ is a *PRF* with output range $\mathbb{B}^{[\ell_{\text{PRFexpand}}/8]}$, where $2^{\ell_{\text{PRFexpand}}}$ is large compared to $r_{\mathbb{J}}$.

Define $f : \mathbb{B}^{[\ell_{sk}]} \times \mathbb{B}^{\mathbb{Y}^{[N]}} \rightarrow \mathbb{F}_{r_{\mathbb{J}}}$ by $f_{sk}(t) := \text{ToScalar}^{\text{Sapling}}(\text{PRF}_{sk}^{\text{expand}}(t))$.

f is also a *PRF* since $\text{LEOS2IP}_{\ell_{\text{PRFexpand}}} : \mathbb{B}^{[\ell_{\text{PRFexpand}}/8]} \rightarrow \{0 \dots 2^{\ell_{\text{PRFexpand}}} - 1\}$ is injective; the bias introduced by reduction modulo $r_{\mathbb{J}}$ is small because § 5.3 ‘*Constants*’ on p. 77 defines $\ell_{\text{PRFexpand}}$ as 512, while $r_{\mathbb{J}}$ has length 252 bits. It follows that the distribution of ask , i.e. $\text{PRF}_{sk}^{\text{expand}}([0]) : sk \xleftarrow{R} \mathbb{B}^{[\ell_{sk}]}$, is computationally indistinguishable from $\text{SpendAuthSig}^{\text{Sapling}}.\text{GenPrivate}()$ defined in § 5.4.7.1 ‘*Spend Authorization Signature (Sapling and Orchard)*’ on p. 98.

- The distribution of nsk , i.e. $\text{ToScalar}^{\text{Sapling}}(\text{PRF}_{sk}^{\text{expand}}([1])) : sk \xleftarrow{R} \mathbb{B}^{[\ell_{sk}]}$, is computationally indistinguishable from the uniform distribution on $\mathbb{F}_{r_{\mathbb{J}}}$. Since $nsk : \mathbb{F}_{r_{\mathbb{J}}} \mapsto \text{repr}_{\mathbb{J}}([nsk] \mathcal{H}^{\text{Sapling}} : \mathbb{J}_{\star}^{(r)})$ is bijective, the distribution of $\text{repr}_{\mathbb{J}}(nsk)$ will be computationally indistinguishable from uniform on $\mathbb{J}_{\star}^{(r)}$ (the keyspace of $\text{PRF}^{\text{nfSapling}}$).